

## Seamless Short Exposure Mode

The data except this specification conform to those of each product.

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### 1. Product Type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

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### 2. Description

The function which can set shorter exposure than Normal exposure mode is described.

This mode allows seamless settings from the normal exposure range to the short exposure range without going through standby state.

\* There is possible that this mode affects the quality of image. Please use this after evaluating and checking enough.

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Application circuits shown, if any, are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits.

### 3. Features

- ◆ Short Exposure mode
  - Seamless Ultra Short Exposure Mode
- ◆ Support mode
  - Sequential Trigger mode
  - Fast Trigger mode
- ◆ Support readout drive mode
  - All-pixel scan mode
  - Vertical / Horizontal 1/2 subsampling mode
  - 2 × 2 FD binning mode (monochrome only)
  - ROI mode
  - Overlap ROI mode
- ◆ Support interface

(IMX530 series / IMX535 series)

| Interface                                                                          |
|------------------------------------------------------------------------------------|
| SLVS output ( 4 ch / 8 ch, 8-bit / 10-bit / 12-bit output)                         |
| SLVS-EC output (1 Lane / 2 Lane / 4 Lane / 8 Lane, 8-bit / 10-bit / 12-bit output) |

(IMX540 series / IMX545 series)

| Interface                                                        |
|------------------------------------------------------------------|
| SLVS output (2 ch / 4 ch / 8 ch, 8-bit / 10-bit / 12-bit output) |
| SLVS-EC output (1 Lane / 2 Lane, 8-bit / 10-bit / 12-bit output) |

(IMX548 series)

| Interface                                                        |
|------------------------------------------------------------------|
| SLVS output (2 ch / 4 ch / 8 ch, 8-bit / 10-bit / 12-bit output) |

(IMX565 series /IMX568 series)

| Interface                                                                  |
|----------------------------------------------------------------------------|
| CSI-2 serial data output (2 Lane / 4 Lane, 8-bit / 10-bit / 12-bit output) |

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## 4. Register Map

The register map for Seamless Ultra Short Exposure Mode is shown below.  
Please refer to the product specification for register setup other than those list.

### 4.1. Register list of Seamless Ultra Short Exposure Mode

#### 4.1.1. Common setting

| Chip ID | Address | Bit   | Register name | Setting value                 |             | Default Value after reset |                  |                  | Reflection timing |
|---------|---------|-------|---------------|-------------------------------|-------------|---------------------------|------------------|------------------|-------------------|
|         |         |       |               | Seamless Ultra Short Exposure | Other modes | IMX530 series *1          | IMX535 series *2 | IMX548 series *3 |                   |
| 13h     | 7Ah     | [7:0] | —             | 01h (1d)                      | A2h (162d)  | D7h (215d)                | A2h (162d)       | A2h (162d)       | S                 |
|         | 7Bh     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | 7Ch     | [7:0] | —             | 16h (22d)                     | 5Bh (91d)   | 5Bh (91d)                 | 5Bh (91d)        | 5Bh (91d)        | S                 |
|         | 7Dh     | [7:0] | —             | 00h (0d)                      | 01h (0d)    | 01h (1d)                  | 01h (1d)         | 01h (1d)         | S                 |
| 10h     | 8Eh     | [7:0] | —             | 02h (2d)                      | A3h (163d)  | D8h (216d)                | A3h (163d)       | A3h (163d)       | S                 |
|         | 8Fh     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | 90h     | [7:0] | —             | 15h (21d)                     | DBh (219d)  | FBh (251d)                | DBh (219d)       | DBh (219d)       | S                 |
|         | 91h     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | BEh     | [7:0] | —             | FFh (255d)                    | DCh (220d)  | FCh (252d)                | DCh (220d)       | DCh (220d)       | S                 |
|         | BFh     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | C0h     | [7:0] | —             | FFh (255d)                    | 5Ah (90d)   | 5Ah (90d)                 | 5Ah (90d)        | 5Ah (90d)        | S                 |
|         | C1h     | [7:0] | —             | 0Fh (15d)                     | 01h (1d)    | 01h (1d)                  | 01h (1d)         | 01h (1d)         | S                 |
| 13h     | BAh     | [7:0] | —             | 01h (1d)                      | 1Bh (27d)   | 75h (117d)                | 1Bh (27d)        | 1Bh (27d)        | S                 |
|         | BBh     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | BCh     | [7:0] | —             | 23h (35d)                     | 5Ch (92d)   | 91h (145d)                | 5Ch (92d)        | 5Ch (92d)        | S                 |
|         | BDh     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
| 11h     | A2h     | [7:0] | —             | 02h (2d)                      | 1Ch (28d)   | 76h (118d)                | 1Ch (28d)        | 1Ch (28d)        | S                 |
|         | A3h     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | A4h     | [7:0] | —             | 22h (34d)                     | 5Bh (91d)   | 90h (144d)                | 5Bh (91d)        | 5Bh (91d)        | S                 |
|         | A5h     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
| 14h     | 0Eh     | [7:0] | —             | 01h (1d)                      | 1Bh (27d)   | 75h (117d)                | 1Bh (27d)        | 1Bh (27d)        | S                 |
|         | 0Fh     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |

| Chip ID | Address | Bit   | Register name | Setting value                 |             | Default Value after reset |                  |                  | Reflection timing |
|---------|---------|-------|---------------|-------------------------------|-------------|---------------------------|------------------|------------------|-------------------|
|         |         |       |               | Seamless Ultra Short Exposure | Other modes | IMX530 series *1          | IMX535 series *2 | IMX548 series *3 |                   |
| 14h     | 10h     | [7:0] | —             | 24h (36d)                     | 70h (112d)  | A5h (165d)                | 70h (112d)       | 70h (112d)       | S                 |
|         | 11h     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
| 12h     | B6h     | [7:0] | —             | 02h (2d)                      | 1Ch (28d)   | 76h (118d)                | 1Ch (28d)        | 1Ch (28d)        | S                 |
|         | B7h     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | B8h     | [7:0] | —             | 23h (35h)                     | 6Fh (111d)  | A4h (164d)                | 6Fh (111d)       | 6Fh (111d)       | S                 |
|         | B9h     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
| 13h     | BEh     | [7:0] | —             | FFh (255d)                    | A2h (162d)  | D7h (215d)                | A2h (162d)       | A2h (162d)       | S                 |
|         | BFh     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | C0h     | [7:0] | —             | FFh (255d)                    | BAh (186d)  | BAh (186d)                | BAh (186d)       | BAh (186d)       | S                 |
|         | C1h     | [7:0] | —             | 0Fh (15d)                     | 01h (1d)    | 01h (1d)                  | 01h (1d)         | 01h (1d)         | S                 |
| 11h     | A6h     | [7:0] | —             | FFh (255d)                    | A3h (163d)  | D8h (216d)                | A3h (163d)       | A3h (163d)       | S                 |
|         | A7h     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | A8h     | [7:0] | —             | FFh (255d)                    | B9h (185d)  | B9h (185d)                | B9h (185d)       | B9h (185d)       | S                 |
|         | A9h     | [7:0] | —             | 0Fh (15d)                     | 01h (1d)    | 01h (1d)                  | 01h (1d)         | 01h (1d)         | S                 |
| 13h     | EAh     | [7:0] | —             | FFh (255d)                    | A2h (162d)  | D7h (215d)                | A2h (162d)       | A2h (162d)       | S                 |
|         | EBh     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | ECh     | [7:0] | —             | FFh (255d)                    | D5h (213d)  | D5h (213d)                | D5h (213d)       | D5h (213d)       | S                 |
|         | EDh     | [7:0] | —             | 0Fh (15d)                     | 01h (1d)    | 01h (1d)                  | 01h (1d)         | 01h (1d)         | S                 |
| 12h     | 46h     | [7:0] | —             | FFh (255d)                    | A3h (163d)  | D8h (216d)                | A3h (163d)       | A3h (163d)       | S                 |
|         | 47h     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | 48h     | [7:0] | —             | FFh (255d)                    | D4h (212d)  | D4h (212d)                | D4h (212d)       | D4h (212d)       | S                 |
|         | 49h     | [7:0] | —             | 0Fh (15d)                     | 01h (1d)    | 01h (1d)                  | 01h (1d)         | 01h (1d)         | S                 |
| 13h     | 9Ah     | [7:0] | —             | 25h (37d)                     | FFh (255d)  | FFh (255d)                | FFh (255d)       | FFh (255d)       | S                 |
|         | 9Bh     | [7:0] | —             | 00h (0d)                      | 0Fh (15d)   | 0Fh (15d)                 | 0Fh (15d)        | 0Fh (15d)        | S                 |
|         | 9Ch     | [7:0] | —             | 3Ah (58d)                     | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | 9Dh     | [7:0] | —             | 00h (0d)                      | 00h (0d)    | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |

| Chip ID | Address | Bit   | Register name | Setting value                 |                                 | Default Value after reset |                  |                  | Reflection timing |
|---------|---------|-------|---------------|-------------------------------|---------------------------------|---------------------------|------------------|------------------|-------------------|
|         |         |       |               | Seamless Ultra Short Exposure | Other modes                     | IMX530 series *1          | IMX535 series *2 | IMX548 series *3 |                   |
| 11h     | 32h     | [7:0] | —             | 26h (38d)                     | FFh (255d)                      | FFh (255d)                | FFh (255d)       | FFh (255d)       | S                 |
|         | 33h     | [7:0] | —             | 00h (0d)                      | 0Fh (15d)                       | 0Fh (15d)                 | 0Fh (15d)        | 0Fh (15d)        | S                 |
|         | 34h     | [7:0] | —             | 39h (57d)                     | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | 35h     | [7:0] | —             | 00h (0d)                      | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
| 13h     | 7Eh     | [7:0] | —             | 25h (37d)                     | FFh (255d)                      | FFh (255d)                | FFh (255d)       | FFh (255d)       | S                 |
|         | 7Fh     | [7:0] | —             | 00h (0d)                      | 0Fh (15d)                       | 0Fh (15d)                 | 0Fh (15d)        | 0Fh (15d)        | S                 |
|         | 80h     | [7:0] | —             | 4Bh (75d)                     | FFh (255d)                      | FFh (255d)                | FFh (255d)       | FFh (255d)       | S                 |
|         | 81h     | [7:0] | —             | 00h (0d)                      | 0Fh (15d)                       | 0Fh (15d)                 | 0Fh (15d)        | 0Fh (15d)        | S                 |
| 10h     | 92h     | [7:0] | —             | 26h (38d)                     | FFh (255d)                      | FFh (255d)                | FFh (255d)       | FFh (255d)       | S                 |
|         | 93h     | [7:0] | —             | 00h (0d)                      | 0Fh (15d)                       | 0Fh (15d)                 | 0Fh (15d)        | 0Fh (15d)        | S                 |
|         | 94h     | [7:0] | —             | 4Ah (74d)                     | FFh (255d)                      | FFh (255d)                | FFh (255d)       | FFh (255d)       | S                 |
|         | 95h     | [7:0] | —             | 00h (0d)                      | 0Fh (15d)                       | 0Fh (15d)                 | 0Fh (15d)        | 0Fh (15d)        | S                 |
| 13h     | A2h     | [7:0] | —             | FFh (255d)                    | 03h (3d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | A3h     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | A4h     | [7:0] | —             | FFh (255d)                    | 2Fh (47d)                       | 21h (33d)                 | 2Fh (47d)        | 2Fh (47d)        | S                 |
|         | A5h     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
| 11h     | 3Ah     | [7:0] | —             | FFh (255d)                    | 04h (4d)                        | 01h (1d)                  | 01h (1d)         | 01h (1d)         | S                 |
|         | 3Bh     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | 3Ch     | [7:0] | —             | FFh (255d)                    | 2Eh (46d)                       | 20h (32d)                 | 2Eh (46d)        | 2Eh (46d)        | S                 |
|         | 3Dh     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
| 13h     | 82h     | [7:0] | —             | FFh (255d)                    | 00h (0d)                        | 10h (16d)                 | 0Eh (14d)        | 0Eh (14d)        | S                 |
|         | 83h     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
|         | 84h     | [7:0] | —             | FFh (255d)                    | 38h (56d) *1<br>40h (64d) *2,*3 | 2Ah (42d)                 | 40h (64d)        | 40h (64d)        | S                 |
|         | 85h     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |
| 10h     | 96h     | [7:0] | —             | FFh (255d)                    | 01h (1d)                        | 11h (17d)                 | 0Fh (15d)        | 0Fh (15d)        | S                 |
|         | 97h     | [7:0] | —             | 0Fh (15d)                     | 00h (0d)                        | 00h (0d)                  | 00h (0d)         | 00h (0d)         | S                 |

| Chip ID | Address | Bit   | Register name | Setting value                                               |                                                       | Default Value after reset   |                             |                             | Reflection timing |
|---------|---------|-------|---------------|-------------------------------------------------------------|-------------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------|
|         |         |       |               | Seamless Ultra Short Exposure                               | Other modes                                           | IMX530 series <sup>*1</sup> | IMX535 series <sup>*2</sup> | IMX548 series <sup>*3</sup> |                   |
| 10h     | 98h     | [7:0] | —             | FFh (255d)                                                  | 37h (55d) <sup>*1</sup><br>3Fh (63d) <sup>*2,*3</sup> | 29h (41d)                   | 3Fh (63d)                   | 3Fh (63d)                   | S                 |
|         | 99h     | [7:0] | —             | 0Fh (15d)                                                   | 00h (0d)                                              | 00h (0d)                    | 00h (0d)                    | 00h (0d)                    | S                 |
| 05h     | 08h     | [7:0] | —             | 00h (0d)                                                    | 4Ah (74d)                                             | 34h (52d)                   | 4Ah (74d)                   | 4Ah (74d)                   | S                 |
|         | 09h     | [7:0] | —             | 00h (0d)                                                    | 00h (0d)                                              | 00h (0d)                    | 00h (0d)                    | 00h (0d)                    | S                 |
|         | 0Ah     | [7:0] | —             | FFh (255d)                                                  | 01h (1d)                                              | 01h (1d)                    | 01h (1d)                    | 01h (1d)                    | S                 |
|         | 0Bh     | [7:0] | —             | 0Fh (15d)                                                   | 00h (0d)                                              | 00h (0d)                    | 00h (0d)                    | 00h (0d)                    | S                 |
| 04h     | ACh     | [7:0] | —             | 01h (1d)                                                    | 4Bh (75d)                                             | 35h (53d)                   | 4Bh (75d)                   | 4Bh (75d)                   | S                 |
|         | ADh     | [7:0] | —             | 00h (0d)                                                    | 00h (0d)                                              | 00h (0d)                    | 00h (0d)                    | 00h (0d)                    | S                 |
|         | A Eh    | [7:0] | —             | FFh (255d)                                                  | 00h (0d)                                              | 00h (0d)                    | 00h (0d)                    | 00h (0d)                    | S                 |
|         | AFh     | [7:0] | —             | 0Fh (15d)                                                   | 00h (0d)                                              | 00h (0d)                    | 00h (0d)                    | 00h (0d)                    | S                 |
| 13h     | 22h     | [7:0] | —             | 0Ah (10d)                                                   | 01h (1d)                                              | 01h (1d)                    | 01h (1d)                    | 01h (1d)                    | S                 |
|         | 23h     | [7:0] | —             | 00h (0d)                                                    | 00h (0d)                                              | 00h (0d)                    | 00h (0d)                    | 00h (0d)                    | S                 |
|         | 24h     | [7:0] | —             | 0Bh (11d)                                                   | 02h (2d)                                              | 02h (2d)                    | 02h (2d)                    | 02h (2d)                    | S                 |
|         | 25h     | [7:0] | —             | 00h (0d)                                                    | 00h (0d)                                              | 00h (0d)                    | 00h (0d)                    | 00h (0d)                    | S                 |
| 02h     | E2h     | [7:0] | GMRWT         | 04h: All-pixel<br>08h: 1/2 subsampling and 2 × 2 FD binning | Refer to data sheet                                   | 0Ch (12d)                   | 10h (16d)                   | 06h (6d)                    | S                 |

<sup>\*1</sup> IMX530 series: IMX530-AAMJ, IMX530-AAQJ, IMX531-AAMJ, IMX531-AAQJ, IMX532-AAMJ, IMX532-AAQJ  
IMX540-AAMJ, IMX540-AAQJ, IMX541-AAMJ, IMX541-AAQJ, IMX542-AAMJ, IMX542-AAQJ

<sup>\*2</sup> IMX535 series: IMX535-AAMJ, IMX535-AAQJ, IMX536-AAMJ, IMX536-AAQJ, IMX537-AAMJ, IMX537-AAQJ  
IMX545-AAMJ, IMX545-AAQJ, IMX546-AAMJ, IMX546-AAQJ, IMX547-AAMJ, IMX547-AAQJ  
IMX565-AAMJ, IMX565-AAQJ, IMX566-AAMJ, IMX566-AAQJ, IMX567-AAMJ, IMX567-AAQJ

<sup>\*3</sup> IMX548 series: IMX548-AAMJ, IMX548-AAQJ, IMX568-AAMJ, IMX568-AAQJ

#### 4.1.2. Shutter Control settings for Sequential Trigger mode

| Chip ID | Address | Bit   | Register name  | Setting value                                   |             | Default Value after reset   |                             |                             | Reflection timing |
|---------|---------|-------|----------------|-------------------------------------------------|-------------|-----------------------------|-----------------------------|-----------------------------|-------------------|
|         |         |       |                | Seamless Ultra Short Exposure                   | Other modes | IMX530 series <sup>*1</sup> | IMX535 series <sup>*2</sup> | IMX548 series <sup>*3</sup> |                   |
| 04h     | B6h     | [7:0] | SHTTIM1 [11:0] | 000h (0d) to 01Ch (28d)                         | 000h (0d)   | 000h (0d)                   | 000h (0d)                   | 000h (0d)                   | V                 |
|         | B7h     | [3:0] |                |                                                 |             |                             |                             |                             |                   |
| 05h     | 12h     | [7:0] | SHTTIM2 [11:0] | 001h (1d) to 01Dh (29d)<br>Set to "SHTTIM1 + 1" | 000h (0d)   | 000h (0d)                   | 000h (0d)                   | 000h (0d)                   | V                 |
|         |         | [3:0] |                |                                                 |             |                             |                             |                             |                   |
|         | 13h     | [7:4] | —              | —                                               | 0h (0d)     | 0h (0d)                     | 0h (0d)                     | 0h (0d)                     | —                 |

<sup>\*1</sup> IMX530 series: IMX530-AAMJ, IMX530-AAQJ, IMX531-AAMJ, IMX531-AAQJ, IMX532-AAMJ, IMX532-AAQJ, IMX540-AAMJ, IMX540-AAQJ, IMX541-AAMJ, IMX541-AAQJ, IMX542-AAMJ, IMX542-AAQJ

<sup>\*2</sup> IMX535 series: IMX535-AAMJ, IMX535-AAQJ, IMX536-AAMJ, IMX536-AAQJ, IMX537-AAMJ, IMX537-AAQJ, IMX545-AAMJ, IMX545-AAQJ, IMX546-AAMJ, IMX546-AAQJ, IMX547-AAMJ, IMX547-AAQJ, IMX565-AAMJ, IMX565-AAQJ, IMX566-AAMJ, IMX566-AAQJ, IMX567-AAMJ, IMX567-AAQJ

<sup>\*3</sup> IMX548 series: IMX548-AAMJ, IMX548-AAQJ, IMX568-AAMJ, IMX568-AAQJ

#### 4.1.3. Dedicated settings for IMX530 series

When using the IMX530 series, perform this setting in addition to the settings in 4.1.1 during standby.

| Chip ID | Address | Bit   | Register name | Setting value                 |             | Default Value after reset   |                             |                             | Reflection timing |
|---------|---------|-------|---------------|-------------------------------|-------------|-----------------------------|-----------------------------|-----------------------------|-------------------|
|         |         |       |               | Seamless Ultra Short Exposure | Other modes | IMX530 series <sup>*1</sup> | IMX535 series <sup>*2</sup> | IMX548 series <sup>*3</sup> |                   |
| 14h     | 34h     | [7:0] | —             | 25h (37d)                     | 24h (36d)   | 24h                         | 24h                         | 24h                         | S                 |
|         | 35h     | [7:0] | —             | 22h (34d)                     | 21h (33d)   | 24h                         | 21h                         | 22h                         | S                 |

<sup>\*1</sup> IMX530 series: IMX530-AAMJ, IMX530-AAQJ, IMX531-AAMJ, IMX531-AAQJ, IMX532-AAMJ, IMX532-AAQJ, IMX540-AAMJ, IMX540-AAQJ, IMX541-AAMJ, IMX541-AAQJ, IMX542-AAMJ, IMX542-AAQJ

<sup>\*2</sup> IMX535 series: IMX535-AAMJ, IMX535-AAQJ, IMX536-AAMJ, IMX536-AAQJ, IMX537-AAMJ, IMX537-AAQJ, IMX545-AAMJ, IMX545-AAQJ, IMX546-AAMJ, IMX546-AAQJ, IMX547-AAMJ, IMX547-AAQJ, IMX565-AAMJ, IMX565-AAQJ, IMX566-AAMJ, IMX566-AAQJ, IMX567-AAMJ, IMX567-AAQJ

<sup>\*3</sup> IMX548 series: IMX548-AAMJ, IMX548-AAQJ, IMX568-AAMJ, IMX568-AAQJ

#### 4.1.4. Dedicated settings for IMX548 series

When using the IMX548 series, perform this setting in addition to the settings in 4.1.1 during standby.

| Chip ID | Address | Bit   | Register name | Setting value                 |             | Default Value after reset   |                             |                             | Reflection timing |
|---------|---------|-------|---------------|-------------------------------|-------------|-----------------------------|-----------------------------|-----------------------------|-------------------|
|         |         |       |               | Seamless Ultra Short Exposure | Other modes | IMX530 series <sup>*1</sup> | IMX535 series <sup>*2</sup> | IMX548 series <sup>*3</sup> |                   |
| 14h     | 33h     | [7:0] | —             | 0Ch (12d)                     | 24h (36d)   | 24h                         | 24h                         | 24h                         | S                 |
|         | 35h     | [7:0] | —             | 21h (33d)                     | 22h (34d)   | 24h                         | 21h                         | 22h                         | S                 |

<sup>\*1</sup> IMX530 series: IMX530-AAMJ, IMX530-AAQJ, IMX531-AAMJ, IMX531-AAQJ, IMX532-AAMJ, IMX532-AAQJ, IMX540-AAMJ, IMX540-AAQJ, IMX541-AAMJ, IMX541-AAQJ, IMX542-AAMJ, IMX542-AAQJ

<sup>\*2</sup> IMX535 series: IMX535-AAMJ, IMX535-AAQJ, IMX536-AAMJ, IMX536-AAQJ, IMX537-AAMJ, IMX537-AAQJ, IMX545-AAMJ, IMX545-AAQJ, IMX546-AAMJ, IMX546-AAQJ, IMX547-AAMJ, IMX547-AAQJ, IMX565-AAMJ, IMX565-AAQJ, IMX566-AAMJ, IMX566-AAQJ, IMX567-AAMJ, IMX567-AAQJ

<sup>\*3</sup> IMX548 series: IMX548-AAMJ, IMX548-AAQJ, IMX568-AAMJ, IMX568-AAQJ

## 5. Description of Seamless Ultra Short Exposure Mode

Seamless Ultra Short Exposure Mode is as below.

◆ Seamless Ultra Short Exposure Mode:

In this mode, you can seamlessly control the exposure time from normal to short exposure time by setting a dedicated register.

Each mode has the characteristics as below.

| Function mode                 | Control method          | Exposure method       | Unit of exposure time | note                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|-------------------------|-----------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seamless Ultra Short Exposure | Sequential Trigger mode | Specialized registers | 53.87 ns              | <ul style="list-style-type: none"> <li>• Minimum exposure time: 1.56 [μs]</li> <li>• The exposure time in the ultra short exposure band is controlled by a dedicated register, while the exposure time in the normal exposure band is controlled by the low level width of XTRIG.</li> <li>• There is possible to affect the image</li> <li>• The ultra short exposure band is affected by the environment.</li> </ul> |
|                               | Fast Trigger mode       | Specialized registers | XTRIG width           | <ul style="list-style-type: none"> <li>• Minimum exposure time: 3.17 [μs]</li> <li>• The exposure time for both the ultra-short exposure band and the normal exposure band is controlled by the low level width of XTRIG.</li> <li>• There is possible to affect the image</li> <li>• The ultra short exposure band is affected by the environment.</li> </ul>                                                         |

## 6. Ultra Short Exposure mode

In this mode, you can seamlessly control the exposure time from normal to short exposure time by setting a dedicated register. But there is possible that this mode affects the quality of image. Please use this after evaluating and checking enough.

There is possible that the exposure time of this mode is changed about  $\pm 500$  ns by the sample and environment (temperature, power voltage and so on). And in case of using the intense light source, the exposure time looks like it is changed by PLS and so on. Please use this after evaluating and checking enough.

### 6.1. Global Shutter (Sequential Trigger Mode) Operation

The exposure time in the ultra short exposure band is controlled by a dedicated register, while the exposure time in the normal exposure band is controlled by the low level width of XTRIG.

#### (1) XTRIG low level pulse width $\geq 5XHS$ (Normal exposure band)

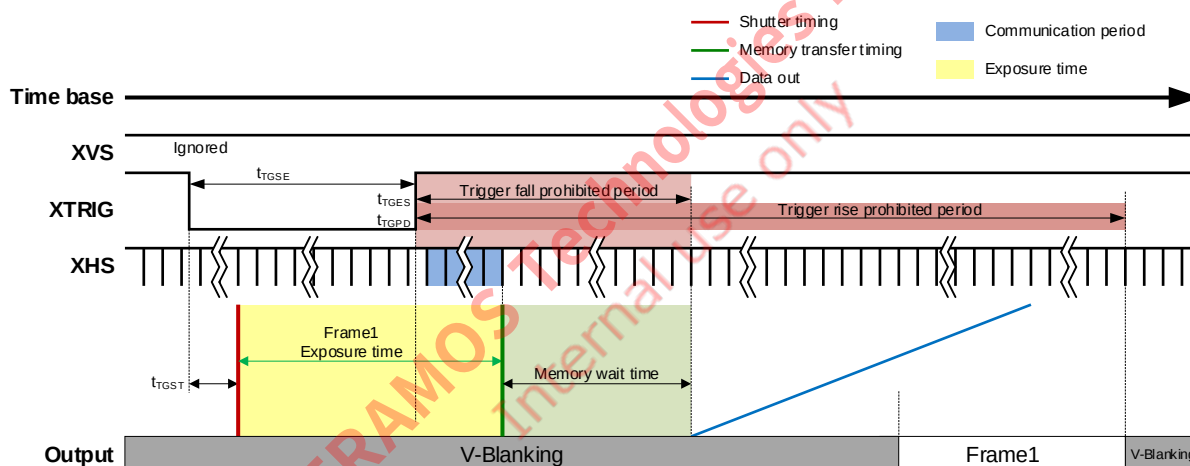
The exposure time is controlled by the low level width of XTRIG.

Set the SHTTIM1 register to 0 and the SHTTIM2 register to 1.

#### Calculation Formula of Exposure Time

$$\text{Exposure time } [\mu\text{s}] = (1 \text{ H period } [\mu\text{s}]) \times (\text{XTRIG low level pulse width } [H]) + 3.13 [\mu\text{s}]^{*1}$$

<sup>\*1</sup> Exposure time error ( $t_{\text{OFFSET}}$ )



### Parameter List of Global Shutter (Sequential Trigger Mode) for all types of image sensors

(Please refer to the product specification for register setup other than those list.)

(This parameter of ROI mode of this section is written in case of all-pixel scan mode in each datasheet as an example.)

| Item                                       | Symbol            | Color / Monochrome | Min.                          | Typ. | Max. | Unit |
|--------------------------------------------|-------------------|--------------------|-------------------------------|------|------|------|
| Integration start delay (All-pixel / ROI)  | $t_{\text{TGST}}$ | Color / Monochrome | $8 [H] + 1.81 [\mu\text{s}]$  | —    | —    | —    |
| Integration start delay (1/2 Subsampling)  |                   | Color / Monochrome | $12 [H] + 1.81 [\mu\text{s}]$ |      | —    |      |
| Integration start delay (2 × 2 FD binning) |                   | Monochrome         | $12 [H] + 1.81 [\mu\text{s}]$ |      | —    |      |
| Pulse width (All-pixel / ROI)              | $t_{\text{TGSE}}$ | Color / Monochrome | 4                             | —    | —    | H    |
| Pulse width (1/2 Subsampling)              |                   | Color / Monochrome | 8                             | —    | —    |      |
| Pulse width (2 × 2 FD binning)             |                   | Monochrome         | 8                             | —    | —    |      |

(2) XTRIG low level pulse width = 4XHS (minimum settings) (Ultra short exposure band)

The exposure time is controlled by the SHTTIM1 and SHTTIM2 registers.

The low level width of XTRIG should be input as 4XHS.

The SHTTIM1 register can be set in the range of 0 (d) to 28 (d). And the SHTTIM2 register should be set to a value that is "SHTTIM1 + 1".

**Calculation Formula of Exposure Time**

$$\text{Exposure time } [\mu\text{s}] = 3.13 [\mu\text{s}]^{*1} - (\text{SHTTIM2} \times 0.05387 [\mu\text{s}])$$

\*1 Exposure time error ( $t_{\text{OFFSET}}$ )

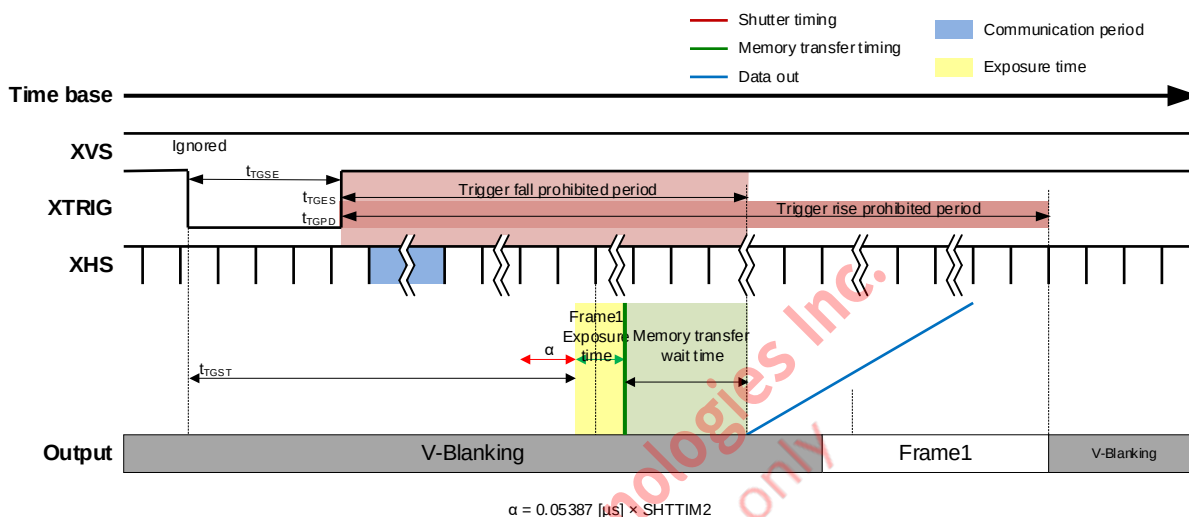


Image Drawing of Global Shutter (Sequential Trigger Mode)

**Parameter List of Global Shutter (Sequential Trigger Mode) for all types of image sensors**

(Please refer to the product specification for register setup other than those list.)

(This parameter of ROI mode of this section is written in case of all-pixel scan mode in each datasheet as an example.)

| Item                                       | Symbol            | Color / Monochrome | Min.                            | Typ. | Max.                            | Unit |
|--------------------------------------------|-------------------|--------------------|---------------------------------|------|---------------------------------|------|
| Integration start delay (All-pixel / ROI)  | $t_{\text{TGST}}$ | Color / Monochrome | 8 [H] + 1.81 [ $\mu\text{s}$ ]  | —    | 8 [H] + 3.32 [ $\mu\text{s}$ ]  | —    |
| Integration start delay (1/2 Subsampling)  |                   | Color / Monochrome | 12 [H] + 1.81 [ $\mu\text{s}$ ] |      | 12 [H] + 3.32 [ $\mu\text{s}$ ] |      |
| Integration start delay (2 × 2 FD binning) |                   | Monochrome         | 12 [H] + 1.81 [ $\mu\text{s}$ ] |      | 12 [H] + 3.32 [ $\mu\text{s}$ ] |      |
| Pulse width (All-pixel / ROI)              | $t_{\text{TGSE}}$ | Color / Monochrome | 4                               | —    | —                               | H    |
| Pulse width (1/2 Subsampling)              |                   | Color / Monochrome | 8                               | —    | —                               |      |
| Pulse width (2 × 2 FD binning)             |                   | Monochrome         | 8                               | —    | —                               |      |

## 6.2. Global Shutter (Fast Trigger Mode) Operation

The exposure time is controlled by the low level width of XTRIG.

### Calculation Formula of Exposure Time

Exposure time [ $\mu\text{s}$ ] = (XTRIG low level pulse width [ $\mu\text{s}$ ] + 3.13 [ $\mu\text{s}$ ])<sup>\*1</sup>

<sup>\*1</sup> Exposure time error ( $t_{\text{OFFSET}}$ )

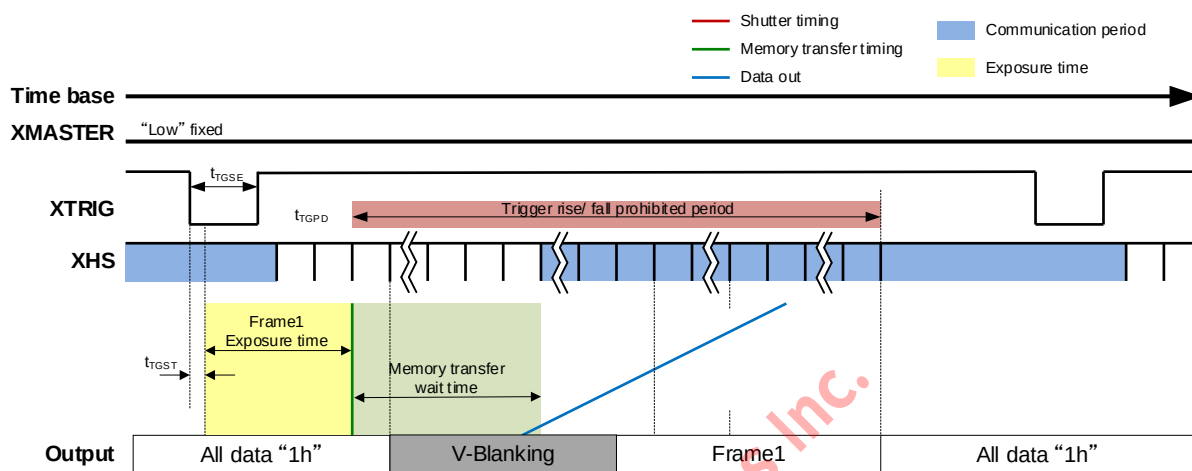


Image Drawing of Global Shutter (Fast Trigger Mode)

### Parameter List of Global Shutter (Fast Trigger Mode) for all types of image sensors

(Please refer to the product specification for register setup other than those list.)

| Item                    | Symbol            | Min. | Typ. | Max. | Unit          |
|-------------------------|-------------------|------|------|------|---------------|
| Integration start delay | $t_{\text{TGST}}$ | —    | —    | 1.84 | $\mu\text{s}$ |
| Pulse width             | $t_{\text{TGSE}}$ | 0.04 | —    | —    | $\mu\text{s}$ |

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## 7. Revision History

| Version | Date           | Page | Remarks       |
|---------|----------------|------|---------------|
| Rev.1.0 | 2024 / 06 / 08 | –    | First Edition |

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